Heuristic-guided reinforcement learning on toy combinatorial problems

Luis Felipe Cattelan

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Abstract

We compare eleven ways of using a hand-written heuristic policy to help PPO, from behaviour cloning with and without a KL anchor, through DAgger, to three methods from the literature: HuRL (heuristic value as reward shaping), HEPO (heuristic reward with a policy-improvement constraint) and HUBL (heuristic blending in offline RL, here IQL). All are evaluated on four toy problems with known heuristics: online bin packing (with two different guides), online knapsack, load balancing on heterogeneous servers and lost-sales inventory control, with 5 seeds and 1.5M environment steps each. We ask whether a method (1) ends better than RL from scratch, (2) reaches the heuristic's level faster than RL from scratch, and (3) ends better than the heuristic. With this budget the answer to (1) is almost always no: only IQL $\rightarrow$ PPO + KL (bin packing with the strong guide) and HuRL (load balancing) are significantly better than PPO from scratch, and warm starts are sometimes significantly worse. The answer to (2) is yes for nearly every method that uses the heuristic offline (IQL/HUBL and DAgger in all five experiments), as long as the guide is beatable. The answer to (3) is yes for every warm-started method in every experiment, but PPO from scratch also beats the heuristic in four of the five experiments.

1 Setting

Questions. Given a decent heuristic policy π_H for a sequential decision problem, which ways of feeding it into reinforcement learning

Q1 end **better than RL from scratch** (final metric of the trained policy, same environment budget)?

Q2 **reach the heuristic's level faster** than RL from scratch (environment steps until the policy first beats the heuristic, counting the heuristic's demonstration steps)?

Q3 end **better than the heuristic** itself?

Problems. Four problems with fixed-length episodes, a discrete (masked) action space, a small vector observation and a per-episode metric where lower is better (Table 1). Each has a *guide* heuristic that the methods learn from and, where we know one, a stronger *reference* heuristic that shows how much room there is above the guide. Bin packing is run twice: guided by the weak Best Fit and by the near-optimal Sum of Squares.

2 Methods

All methods share the same PPO: separate 2×128 tanh MLPs for actor and critic, 16 parallel environments × 128 steps per update, 4 epochs, minibatch 512, Adam $3 \cdot 10^{-4}$, $\gamma = 1$ (finite horizon), GAE $\lambda = 0.95$, clip 0.2, entropy 0.01, value coefficient 0.5, gradient norm 0.5. The heuristic is a policy $\pi_H(s)$, so the literature methods built around a heuristic reward or value were adapted as described.

Problem	Decision (episode)	Metric	Guide	Reference
Online bin packing	place an item of size 3 or 4 ($p = 2/3, 1/3$) into a bin of capacity 12 (200 items)	wasted capacity	Best Fit 9.68% / Sum of Squares 2.36%	Sum of Squares 2.36%
Online knapsack	accept/reject items, weight and value uniform on {1..10}, capacity 250 (200 items)	gap to the offline fractional optimum	static density threshold 3.69%	adaptive threshold 1.24%
Load balancing	send a job (size $\sim$Exp(1)) to one of 4 FIFO servers with speeds 0.4–1.6, load 0.85 (200 jobs)	mean sojourn time	shortest expected delay 1.73	— (least work left: 1.96)
Lost-sales inventory	order 0–15 units, lead time 3, Poisson(5) demand, holding 1, lost sale 9 (100 periods)	mean cost per period	base stock ($S = 23$) 6.94	capped base stock 6.70

Table 1: The toy problems. Heuristic values are on 256 fixed evaluation episodes. A uniformly random policy scores 13.6%, 55.0%, 3.99 and 129.0 respectively.

1. **PPO from scratch.** The heuristic is not used.
2. **BC $\rightarrow$ PPO (naive).** Behaviour cloning on 64 heuristic episodes (cross-entropy, 10 epochs), then PPO from the cloned policy.
3. **BC $\rightarrow$ PPO + KL.** Same, but PPO adds $\beta_u \,\mathrm{KL}(\pi \,\|\, \pi_{\mathrm{BC}})$ to the frozen clone, with β decaying linearly from 1 to 0 by 50% (or 20%) of training, and the first 10 updates train only the critic.
4. **DAgger $\rightarrow$ PPO (naive / + KL).** The same 64-episode budget in 4 rounds: round 0 is the heuristic's own play, later rounds let the current policy act while the heuristic labels the visited states; retrain on the aggregate after each round. Then PPO as in 2 or 3.
5. **HuRL** (Cheng et al., 2021). A heuristic value $h(s)$ is fitted by least squares to the heuristic's Monte-Carlo returns on the demonstrations. PPO then trains on the reshaped problem $\tilde{r} = r + (1 - \lambda_n)\gamma\, h(s')$, $\tilde{\gamma} = \lambda_n \gamma$, with $\lambda_n = \lambda_0 + (1 - \lambda_0)\tanh(\omega n)$, $\lambda_0 = 0.9$, reaching ≈ 1 at half of training. Run from scratch and from the BC policy (**BC $\rightarrow$ HuRL**).
6. **HEPO** (Lee et al., 2024). Maximise $(1 + \alpha)J(\pi) + H(\pi)$ subject to $J(\pi) \geq J(\pi_H)$. Adaptation: the heuristic reward is $h(s, a) = 0.1 \cdot \mathbb{1}[a = \pi_H(s)]$, whose optimal policy is the heuristic itself, so π_H is the heuristic (not trained). Half of the environments are played by π_H on the same random sequences; π is updated on both halves with advantage $(1 + \alpha)A_r + A_h$ and $\pi_H(a \mid s) = 1$ in the importance ratio; $\alpha \leftarrow \max(0, \alpha - (\hat{J}(\pi) - \hat{J}(\pi_H)))$ from Monte-Carlo returns.
7. **IQL $\rightarrow$ PPO + KL.** Offline RL on 64 ε-greedy heuristic episodes with $\varepsilon \in [0, 0.5]$ (mixed quality): discrete IQL (expectile 0.7, $\gamma = 0.99$, 10k updates) with an advantage-weighted actor ($\beta = 10$); then PPO + KL to that actor.
8. **IQL + HUBL $\rightarrow$ PPO + KL** (Geng et al., 2023). Same, with HUBL's relabelled targets $\tilde{r} = r + \gamma\lambda h'$, $\tilde{\gamma} = \gamma(1 - \lambda)$, h the Monte-Carlo return-to-go and $\lambda = 0.1 \cdot \mathrm{rank}(\tau)/n$ per trajectory.

Protocol. 5 seeds per method and experiment, 1.5M PPO steps. The heuristic's offline data (64 episodes, i.e. 12,800 steps; 6,400 for inventory) is counted as environment interaction; HEPO's heuristic half is part of its PPO steps. Policies are evaluated greedily every 10 updates on 64 fixed episodes (learning curves) and at the end on 256 fixed episodes, the same for every method and heuristic. Method-specific hyperparameters were set once from the papers' recommendations and not tuned.

Method	Bin packing (Best Fit)	Bin packing (Sum of Sq.)	Knapsack (static thr.)	Load bal. (SED)	Inventory (base stock)	Q1	Q2	Q3
PPO from scratch	– – ✓	– – ≈	– – ✓	– – ✓	– – ✓	–	–	4/5
BC → PPO (naive)	≈ ✗ ✓	≈ ✓ ✓	≈ ✓ ✓	≈ ✓ ✓	≈ ✓ ✓	0/5	4/5	5/5
BC → PPO + KL (50%)	≈ ✗ ✓	≈ ✓ ✓	✗ ✓ ✓	✗ ✓ ✓	✗ ✓ ✓	0/5	4/5	5/5
BC → PPO + KL (20%)	≈ ✗ ✓	≈ ✓ ✓	✗ ✓ ✓	≈ ✓ ✓	≈ ✓ ✓	0/5	4/5	5/5
DAgger → PPO (naive)	≈ ✓ ✓	≈ ✓ ✓	✗ ✓ ✓	✗ ✓ ✓	≈ ✓ ✓	0/5	5/5	5/5
DAgger → PPO + KL (50%)	≈ ✗ ✓	≈ ✓ ✓	✗ ✗ ✓	✗ ✓ ✓	≈ ✓ ✓	0/5	3/5	5/5
HuRL (from scratch)	≈ ✓ ✓	≈ ✗ ≈	✗ ✓ ✓	✓ ✗ ✓	≈ ✗ ✓	1/5	2/5	4/5
BC → HuRL	≈ ✓ ✓	≈ ✓ ✓	✗ ✓ ✓	≈ ✓ ✓	≈ ✓ ✓	0/5	5/5	5/5
HEPO (from scratch)	✗ ✓ ≈	≈ ✓ ≈	✗ ✓ ✓	✗ ✓ ✓	✗ ✗ ✓	0/5	4/5	3/5
IQL → PPO + KL	≈ ✓ ✓	✓ ✓ ✓	≈ ✓ ✓	✗ ✓ ✓	≈ ✓ ✓	1/5	5/5	5/5
IQL + HUBL → PPO + KL	≈ ✓ ✓	≈ ✓ ✓	≈ ✓ ✓	✗ ✓ ✓	✗ ✓ ✓	0/5	5/5	5/5

Table 2: Scorecard for the three questions. Each cell gives Q1 Q2 Q3 for one experiment: Q1 final metric better than PPO from scratch, Q2 beats the guide heuristic in fewer environment steps than PPO from scratch, Q3 final metric better than the guide heuristic. ✓ yes, ✗ no (significantly worse, slower or never), ≈ no significant difference. Q1 and Q3 use a one-sided Welch t-test at 5% over the 5 seeds (Q3 against the heuristic's value); Q2 compares the median steps of Table 4. Last columns: number of experiments with ✓.

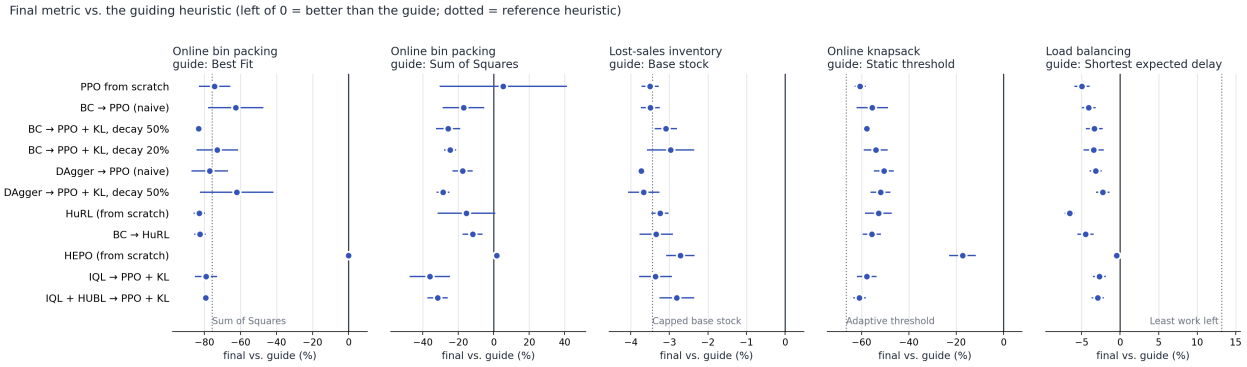


Figure 1: Final metric relative to the guide heuristic (mean ± std over seeds; left of 0 is better than the guide). Dotted: the reference heuristic when it is within the plotted range.

3 Results

Table 2 answers the three questions per method and experiment; Table 3 gives the final metric (Q1, Q3) and Table 4 the steps to beat the guide (Q2). Figures 1 and 2 show the final metric relative to the guide and the learning curves.

Q1: better than RL from scratch? Rarely, with 1.5M steps.

- Only two cases are significantly better than PPO from scratch: IQL → PPO + KL on bin packing with Sum of Squares (1.51% vs 2.48%) and HuRL on load balancing (1.62 vs 1.64).
- Most other cases are within noise. On bin packing PPO from scratch varies a lot between seeds (±0.85%), so even large mean gains (BC → PPO + KL 1.63% vs 2.48% with Best Fit) are not significant with 5 seeds.
- Several are significantly *worse*: on knapsack and load balancing, PPO from scratch ends at 1.45% and 1.64, and most warm starts end at 1.56–1.82% and 1.65–1.69. Starting from the heuristic, and especially being anchored to it, can leave the policy short of what PPO finds alone.
- HEPO is significantly worse than PPO from scratch in four experiments: it ends at the guide's level (Q3 below).

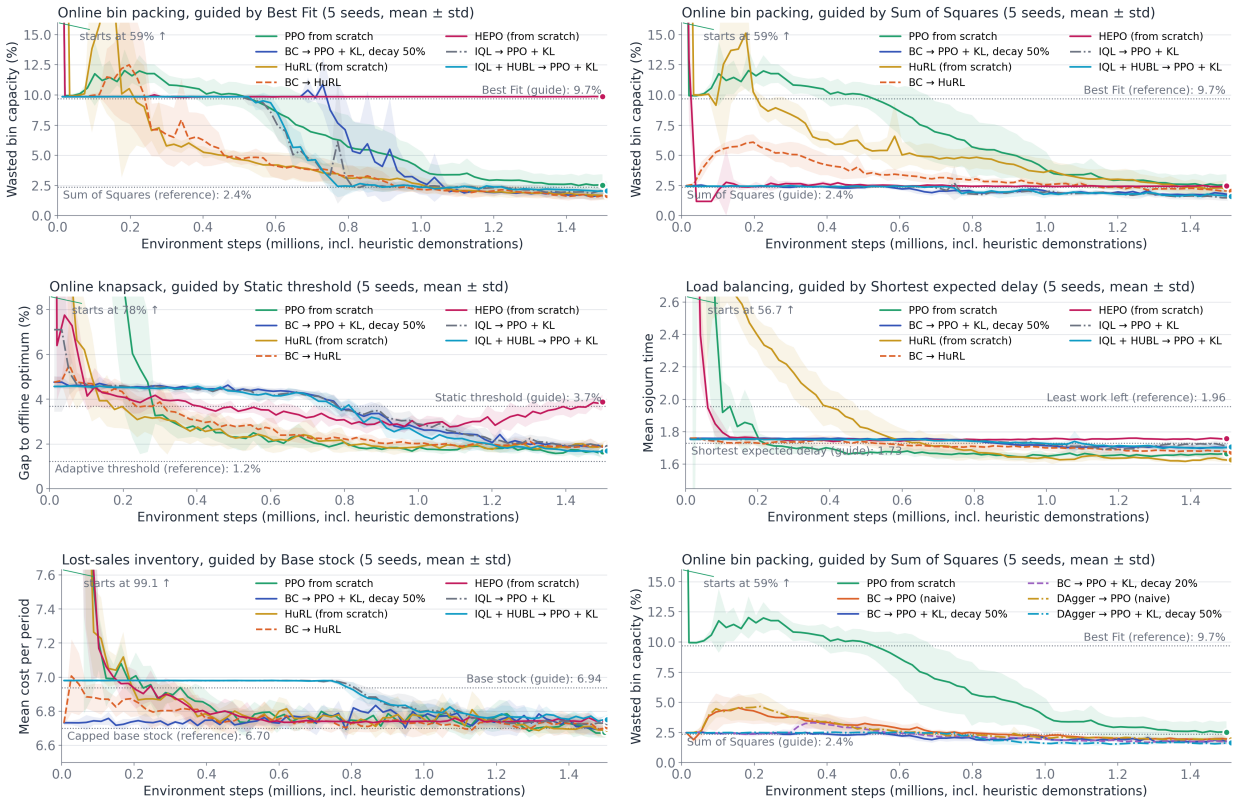


Figure 2: Learning curves (mean ± std over 5 seeds, x-axis includes the heuristic's demonstration steps). First five panels: the literature methods against PPO from scratch and BC → PPO + KL on each experiment. Last panel: the cloning variants (BC/DAgger, naive and with KL) on bin packing guided by Sum of Squares.

Method	Bin packing (Best Fit)	Bin packing (Sum of Sq.)	Knapsack (static thr.)	Load bal. (SED)	Inventory (base stock)
Guide heuristic	9.68	2.36	3.69	1.73	6.94
PPO from scratch	2.48 ±0.85	2.48 ±0.85	1.45 ±0.09	1.64 ±0.02	6.70 ±0.02
BC → PPO (naive)	3.62 ±1.48	1.95 ±0.28	1.64 ±0.25	1.66 ±0.02	6.70 ±0.02
BC → PPO + KL (50%)	**1.63 ±0.23**	1.75 ±0.16	1.56 ±0.06	1.67 ±0.02	6.72 ±0.02
BC → PPO + KL (20%)	2.63 ±1.11	1.78 ±0.08	1.70 ±0.19	1.67 ±0.02	6.73 ±0.04
DAgger → PPO (naive)	2.22 ±0.98	1.94 ±0.14	1.82 ±0.16	1.67 ±0.01	**6.68 ±0.01**
DAgger → PPO + KL (50%)	3.67 ±1.97	1.68 ±0.09	1.77 ±0.15	1.69 ±0.02	6.68 ±0.03
HuRL (from scratch)	1.66 ±0.31	1.99 ±0.38	1.74 ±0.21	**1.62 ±0.01**	6.71 ±0.02
BC → HuRL	1.70 ±0.31	2.07 ±0.13	1.63 ±0.14	1.65 ±0.02	6.71 ±0.03
HEPO (from scratch)	9.68 ±0.00	2.40 ±0.05	3.05 ±0.21	1.72 ±0.00	6.75 ±0.03
IQL → PPO + KL	2.02 ±0.60	**1.51 ±0.27**	1.55 ±0.15	1.68 ±0.01	6.70 ±0.03
IQL + HUBL → PPO + KL	2.00 ±0.18	1.61 ±0.13	**1.44 ±0.10**	1.68 ±0.01	6.74 ±0.03

Table 3: Final performance of the trained policy (greedy, 256 fixed evaluation episodes; mean ± std over 5 seeds; best per column in bold). Bin packing: wasted capacity (%); knapsack: gap to the offline optimum (%); load balancing: mean sojourn time; inventory: mean cost per period. Lower is better everywhere.

Method	Bin packing (Best Fit)	Bin packing (Sum of Sq.)	Knapsack (static thr.)	Load bal. (SED)	Inventory (base stock)
PPO from scratch	573k	1157k	225k	205k	102k
BC → PPO (naive)	812k	33k	136k	**13k**	**6k**
BC → PPO + KL (50%)	771k	54k	115k	**13k**	**6k**
BC → PPO + KL (20%)	648k	74k	115k	**13k**	**6k**
DAgger → PPO (naive)	443k	54k	177k	**13k**	**6k**
DAgger → PPO + KL (50%)	832k	279k	484k	**13k**	**6k**
HuRL (from scratch)	**156k**	1221k	115k	504k	191k
BC → HuRL	218k	730k	115k	**13k**	**6k**
HEPO (from scratch)	328k	41k	123k	143k	164k
IQL → PPO + KL	566k	**13k**	115k	**13k**	**6k**
IQL + HUBL → PPO + KL	545k	54k	**13k**	**13k**	**6k**

Table 4: Environment steps until the policy first beats its guide heuristic on the 64 evaluation episodes of the learning curves (median over seeds; includes the heuristic's demonstration steps).

Q2: reaches the heuristic's level faster? Yes, for most methods that use the heuristic offline, as long as the guide is beatable.

- IQL → PPO + KL, IQL + HUBL → PPO + KL, DAgger → PPO and BC → HuRL are faster than PPO from scratch in all five experiments; BC → PPO (naive or + KL) in four.
- With a strong guide the gain is large: 13k steps instead of 205k on load balancing, 6k instead of 102k on inventory, and 13–74k for most warm starts instead of 1157k on bin packing with Sum of Squares (where PPO from scratch beats the guide in only 4 of 5 seeds).
- With the weak Best Fit guide, cloning is *slower* (BC → PPO 812k, BC → PPO + KL 771k, vs 573k): the clone starts exactly at Best Fit's level and has to unlearn it, and the KL anchor resists that. Methods that change the learning problem instead are faster: HuRL from scratch 156k, BC → HuRL 218k.
- HuRL from scratch is the least reliable here (faster in two of five): its shortened horizon helps on bin packing with Best Fit and on knapsack, but not on load balancing (504k vs 205k).

Q3: better than the heuristic? Yes, for every warm-started method; but this does not separate the methods from PPO from scratch, which beats the heuristic in four of five experiments too.

- All warm-started methods end significantly better than the guide in all five experiments, and HuRL in four (with Sum of Squares it is within noise: 1.99 ± 0.38% vs 2.36%).
- PPO from scratch also does, except on bin packing with Sum of Squares, where it is within noise of the guide (2.48% vs 2.36%).
- HEPO stays at the guide's level on bin packing (9.68% ± 0.00 with Best Fit, 2.40% with Sum of Squares). Its multiplier α keeps growing because the sampled (exploring) policy rarely beats the deterministic heuristic in our Monte-Carlo estimate, and the heuristic's half of the data keeps pulling π towards π_H. We attribute this to our adaptation rather than to HEPO itself.
- How far beyond: on bin packing the warm starts end 12–36% below Sum of Squares' waste and 62–83% below Best Fit's; on knapsack the best methods halve the static threshold's gap (1.44% vs 3.69%) but none reaches the adaptive threshold (1.24%); on inventory all methods reach the level of capped base stock (6.68–6.75 vs 6.70).

4 Conclusions

- **Q2 is where the heuristic pays off.** Offline use of a strong heuristic (IQL/HUBL, DAgger, BC) passes it 10–20 times sooner than PPO from scratch (load balancing, inventory, bin packing with Sum of Squares); IQL/HUBL → PPO + KL, DAgger → PPO and BC → HuRL are faster in every experiment, though only marginally with the weak Best Fit guide for IQL/HUBL.
- **Q1 is rarely answered yes** with a budget at which PPO from scratch converges: only IQL → PPO + KL (bin packing, strong guide) and HuRL (load balancing) end significantly better, and anchored warm starts can end worse. A smaller budget is the natural test of whether the lead from Q2 turns into a final advantage.
- **Q3 is easy on these problems**: nearly every method, including PPO from scratch, ends beyond the guide; the stronger references (adaptive threshold, capped base stock) are matched at best.
- With a **weak guide**, cloning it does not even speed things up; reward shaping with its value (HuRL) does.
- **HEPO**, as adapted to a heuristic policy, does not get past the guide on bin packing; a fairer version would compare greedy returns in the multiplier update or leave the heuristic's half out of the policy update.

Limitations. 5 seeds (so only large differences are significant), one training budget, untuned method-specific hyperparameters, and small problems whose guides are fully observable functions of the state (so cloning is almost exact). Q2 uses the first evaluation below the guide on 64 episodes, which is noisy for methods that start right at the guide's level.

Reproducing.

- `python run_experiments.py --problem P [--demo H]` trains all methods for one experiment, with `P` one of `binpack`, `knapsack`, `load_balancing`, `inventory`;
- `python plot_results.py` redraws every figure;
- `python report/make_tables.py` regenerates the tables of this report.